



12071 - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI Imaging	MIRI Imaging	(1) M33-GM-1
	2	LRS spectroscopy	MIRI Low Resolution Spectroscopy	(1) M33-GM-1
	3	NIRSpec spectroscopy	NIRSpec Fixed Slit Spectroscopy	(1) M33-GM-1

ABSTRACT

Eruptions from stellar mergers involving giant stars are predicted to be prolific dust factories contributing significantly to the cosmic dust budget. Yet these events have remained largely hidden for decades due to a lack of sensitive infrared searches. The only confirmed giant-star merger erupted over twenty years ago and was discovered only retrospectively in archival data. It was later found to have produced a dust yield rivaling core-collapse supernovae, but its early molecule and dust formation phases were entirely missed.

JWST Proposal 12071 (Created: Tuesday, May 19, 2026, 4:00:17PM Eastern Standard Time) - Overview

We have now discovered the second giant star merger in an ongoing search for mid-IR explosions using the NEOWISE survey. Unlike its predecessor, this source has been caught at an early phase, with recent mid-IR photometry indicating that dust formation has just begun. This presents an unprecedented opportunity to trace molecular and dust formation in a highly dust-productive environment from early times.

Here, we propose a monitoring campaign with JWST in Cycle 5, 6, and 7, combining NIRSpec to characterize molecules such as CO, SiO, water-vapor, and potentially water-ice, MIRI LRS spectroscopy to detect the seeds of oxygen-rich dust, and long-wavelength MIRI imaging to trace the coldest dust components. These observations will trace the evolution of silicate dust formation in a giant star merger, and test models of oxygen-rich dust formation and common-envelope evolution. With dust formation already underway, deferring the initiation of this campaign to future cycles risks entirely missing out on the early dust-forming phase of this likely once-in-a-JWST-lifetime event.

OBSERVING DESCRIPTION

In this proposal, we aim to monitor a newly discovered giant star merger M33-GM-1 - the first in the last two decades and only the second one overall - with the James Webb Space Telescope in Cycle 5, 6, and 7, to study the evolution of its molecular and dust content. Our Cycle 5 observations constitute a total time of 4.7 hours including overheads (1.9 hours science), and comprise -

- 1) NIRSpec spectroscopy : With the high-resolution G235H/F170LP (2828 seconds) and G395H/F290LP (1514 seconds) grating-filter combinations covering 1.6 to 5um
- 2) MIRI LRS spectroscopy covering 5-12um (1676 seconds)
- 3) MIRI imaging : In F1280W, F1500W, F1800W, F2100W (122 seconds each), and F2550W (255 seconds) filters.

Proposal 12071 - Targets - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	M33-GM-1	RA: 01 33 18.9730 (23.3290542d) Dec: +30 52 13.23 (30.87034d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO					

Proposal 12071 - Observation 1 - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Observation	Proposal 12071, Observation 1: MIRI Imaging										Tue May 19 21:00:17 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	M33-GM-1	RA: 01 33 18.9730 (23.3290542d)			Epoch of Position: 2000.0					
			Dec: +30 52 13.23 (30.87034d)								
			Equinox: J2000								
Template	Comments:										
	Category=Star										
	Description=[M stars, Nova-like variable stars, Novae]										
	Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	5	4		1	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1280W	FASTR1	10	1	1	Dither 1	4	4	111.002	277769
	2	F1500W	FASTR1	10	1	1	Dither 1	4	4	111.002	277769
	3	F1800W	FASTR1	10	1	1	Dither 1	4	4	111.002	277769
	4	F2100W	FASTR1	10	1	1	Dither 1	4	4	111.002	277769
	5	F2550W	FASTR1	8	3	1	Dither 1	4	12	288.604	277769
Special Requirements	Group Observations 1, 2, 3, Non-interruptible										

Proposal 12071 - Observation 2 - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Observation	Proposal 12071, Observation 2: LRS spectroscopy									Tue May 19 21:00:17 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	M33-GM-1	RA: 01 33 18.9730 (23.3290542d)			Epoch of Position: 2000.0				
			Dec: +30 52 13.23 (30.87034d)							
			Equinox: J2000							
		Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 M33-GM-1	F560W	FAST	8	1	1	22.2	277769	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	5	1	1	1	1	13.875		F560W

Proposal 12071 - Observation 2 - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	100	3	6	1	2	1676.124	277769
Special Requirements	Group Observations 1, 2, 3, Non-interruptible								

Proposal 12071 - Observation 3 - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Observation	Proposal 12071, Observation 3: NIRSpec spectroscopy											Tue May 19 21:00:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	M33-GM-1	RA: 01 33 18.9730 (23.3290542d)				Epoch of Position: 2000.0					
			Dec: +30 52 13.23 (30.87034d)									
			Equinox: J2000									
		Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 M33-GM-1	WATA	FULL	F110W	NRSRAPIDD6	3	1	1	171.788	277769	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	3							NONE			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	S200A1	NRSIRS2RAPID	32	2	1	NONE	3	6	2888.6	
	2	G395H/F290LP	S200A1	NRSIRS2RAPID	35	1	2	NONE	3	3	1575.6	

Proposal 12071 - Observation 3 - An Elusive Dust Factory at Work: Tracing real-time dust and molecule formation in a giant star merger

Special Requirements	Group Observations 1, 2, 3, Non-interruptible
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